

A No-Go Result for an Infinite Cuntz-Pair Proper-Set Swindle

FA/Banach run `fa.banach.001`

June 13, 2026

Source question

Albiac and Ansorena ask in arXiv:2603.07324, Question 2.15: *Is there a square stable Banach space failing to be hyperplane stable?* The present packet does not answer this question. It records a route no-go: one cannot implement the previously isolated fresh-pair Cuntz-index swindle inside the usual Gowers–Maurey proper-set-of-spreads framework.

By Proposition 2.11, only c_0 , ℓ_1 , and ℓ_2 can be unconditional spreading models of these direct sums. By Proposition 2.12, ℓ_p and ℓ_q are the unique unconditional spreading models of $Z_{p,q}$, $p, q \in [1, \infty]$. In particular, any unconditional spreading model of $c_0(\ell_1)$, $\ell_1(c_0)$, $c_0(\ell_2)$ or $\ell_1(\ell_2)$ is c_0 , ℓ_1 or ℓ_2 .

Those who still expect nice universal structural results for Banach spaces would be tempted to conjecture that the answer to Question 2.8 is positive; however, as we will see in Section 5, this is not the case.

2.4. The hyperplane problem and related questions. Given a Banach space $\mathbb{X}$, there is $h = h(\mathbb{X}) \in \mathbb{N}_0$ such that

$$\{m \in \mathbb{N} : \mathbb{X} \oplus \mathbb{F}^m \simeq \mathbb{X}\} = \{hk : k \in \mathbb{N}\}$$

(see [7, Lemma 2.8]).

Definition 2.13. We say that a Banach space $\mathbb{X}$ is *hyperplane stable* if $h(\mathbb{X}) = 1$, i.e., $\mathbb{X} \oplus \mathbb{F} \simeq \mathbb{X}$. If $h(\mathbb{X}) > 0$, i.e., there is $h \in \mathbb{F}$ such that $\mathbb{X} \simeq \mathbb{X} \oplus \mathbb{F}^h$ we say that $\mathbb{X}$ is *hyperplane-sum stable*.

Similarly, there is $s = s(\mathbb{X})$ such that

$$\{m \in \mathbb{N} : \mathbb{X}^m \simeq \mathbb{X}\} = \{1 + sk : k \in \mathbb{N}\}.$$

Definition 2.14. A Banach space $\mathbb{X}$ is *square stable* if $s(\mathbb{X}) = 1$, i.e., $\mathbb{X}^2 \simeq \mathbb{X}$. If $s(\mathbb{X}) > 0$, i.e., there is $m \in \mathbb{N} \cap [2, \infty)$ such that $\mathbb{X}^m \simeq \mathbb{X}$, we say that $\mathbb{X}$ is *power stable*.

Most spaces that arise naturally in functional analysis are both hyperplane stable and square stable. The first counterexample to this paradigm goes back to [19], where James provided a Banach space $\mathbb{J}$ with $s(\mathbb{J}) = 0$ and $h(\mathbb{J}) = 1$.

The space $\mathbb{G}$ constructed by Gowers [15] witnesses that the answer to the Banach's hyperplane problem is negative, that is, there is a hyperplane nonstable Banach space. In fact, Gowers [15] proved that $h(\mathbb{G}) = 0$ and, later on, Gowers and Maurey [17] proved that $s(\mathbb{G}) = 0$. So, Gowers space $\mathbb{G}$ witnesses the existence of Banach spaces that are neither hyperplane stable nor square stable.

Gowers went on in [16] to build a Banach space $\mathbb{M}$ with $s(\mathbb{M}) = 2$. Subsequently, Gowers and Maurey [17] constructed for each $m \in \mathbb{N}$ a Banach space $\mathbb{S}_m$ with $s(\mathbb{S}_m) = m$, and another Banach space $\mathbb{H}_m$ with $h(\mathbb{H}_m) = m$. In the same paper, Gowers and Maurey conjectured that $h(\mathbb{S}_m) = 0$ for all $m \in \mathbb{N}$. To the best of our knowledge, this problem is still open. Similarly, we do not know whether the spaces $\mathbb{H}_m$, $m \in \mathbb{N}$, are square stable. In fact, it seems that the next question is still open.

Question 2.15. Is there a square stable Banach space failing to be hyperplane stable?

The Schröder–Bernstein's problem for Banach spaces asks whether, assuming $\mathbb{X}$ and $\mathbb{Y}$ are Banach spaces with $\mathbb{X} \triangleleft \mathbb{Y} \triangleleft \mathbb{X}$, then $\mathbb{X} \simeq \mathbb{Y}$.

Spreads and exact binary pairs

We use the standard Gowers–Maurey terminology. A spread is the partial coordinate operator determined by two infinite subsets of $\mathbb{N}$; it maps the increasing enumeration of its domain onto the increasing enumeration of its range and kills the complementary coordinates. In particular, the spread from all of $\mathbb{N}$ onto a fixed infinite set is unique.

Definition 1. A set $\mathcal{S}$ of spreads is *proper* if, for every two distinct matrix positions $(p, q) \neq (r, s)$, only finitely many spreads in $\mathcal{S}$ have nonzero matrix coefficients at both positions.

Definition 2. An exact binary Cuntz pair in $\mathcal{S}$ is a pair of spreads (R_0, R_1) such that

$$R_i^* R_j = \delta_{ij} I, \quad R_0 R_0^* + R_1 R_1^* = I.$$

Writing $A_i = \text{ran } R_i$, the sets A_0, A_1 are disjoint infinite sets with union $\mathbb{N}$, and the range projections $P_{A_i} = R_i R_i^*$ are coordinate-projection spreads in $\mathcal{S}$.

Theorem 1. Let $\mathcal{S}$ be a proper set of spreads. Then $\mathcal{S}$ contains only finitely many exact binary Cuntz pairs. Consequently it cannot contain countably many pairwise commuting exact binary Cuntz pairs.

Proof. Assume first that infinitely many distinct unordered binary partitions

$$\mathbb{N} = A_{m,0} \sqcup A_{m,1}$$

come from exact binary Cuntz pairs in $\mathcal{S}$. Choose three distinct indices $a, b, c \in \mathbb{N}$. For each binary partition, at least one of the three unordered pairs $\{a, b\}$, $\{a, c\}$, $\{b, c\}$ lies in one side of the partition. By the pigeonhole principle, one fixed pair, say $\{a, b\}$, lies in one side for infinitely many of the partitions.

For each of these partitions let B_m be the side containing both a and b . The coordinate projection P_{B_m} is a spread in $\mathcal{S}$, and it has nonzero diagonal coefficients at both distinct matrix positions (a, a) and (b, b) . The sets B_m are distinct: if $B_m = B_n$, then their complements are also equal, so the unordered binary partitions are equal. Thus $\mathcal{S}$ contains infinitely many spreads having nonzero coefficients at the two fixed distinct positions (a, a) and (b, b) , contradicting properness.

Therefore only finitely many unordered binary partitions can occur. For a fixed labeled partition (A_0, A_1) , the spread from all of $\mathbb{N}$ onto A_i is unique, so a fixed unordered partition gives at most two labeled exact binary Cuntz pairs, according to the label swap. Hence only finitely many exact binary Cuntz pairs occur in total. $\square$

Remark 1. The obstruction does not use commutativity. Properness alone forbids the infinite reserve of exact binary decompositions required by the fresh-pair Fredholm-index swindle. Thus any continuation of the square-stable but hyperplane-nonstable program must either complete the single binary-tree Fredholm-index argument, relax exactness/properness, or use a different construction.